We also mention that in many examples, including Examples 7.1.3(1)-(6), the word "class" in part (ii) of Definition 7.1.1 can be replaced by "set". Categories with this property (that $\mathrm{Hom}(X, Y)$ is a set for any X, Y) are called **locally small**; many categories that we encounter are of this kind.

Definition 7.1.4. A **full subcategory** of a category $\mathcal{C}$ is a category $\mathcal{C}'$ whose objects are a subclass of objects of $\mathcal{C}$, and $\mathrm{Hom}_{\mathcal{C}'}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(X, Y)$.

Example 7.1.5. The category **AbelianGroups** is a full subcategory of the category **Groups**.

Sometimes the collection $\mathrm{Hom}(X, Y)$ of morphisms from X to Y in a given locally small category $\mathcal{C}$ is not just a set but has some additional structure (say, the structure of an abelian group, or a vector space over some field). In this case one says that $\mathcal{C}$ is **enriched** over another category $\mathcal{D}$ (which is a **monoidal** category, i.e., has a product operation and a unit object under this product, e.g., the category of abelian groups or vector spaces with the tensor product operation). This means that for each $X, Y \in \mathcal{C}$, $\mathrm{Hom}(X, Y)$ is an object of $\mathcal{D}$, and the composition $\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z)$ is a morphism in $\mathcal{D}$. E.g., if $\mathcal{D}$ is the category of vector spaces, this means that the composition is bilinear, i.e., gives rise to a linear map $\mathrm{Hom}(Y, Z) \otimes \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z)$. For a more detailed discussion of this, we refer the reader to [**McL**].

Example 7.1.6. The category $\mathrm{Rep}(A)$ of representations of a k-algebra A is enriched over the category of k-vector spaces.

7.2. Functors

We would like to define arrows between categories. Such arrows are called functors.

Definition 7.2.1. A **functor** $F : \mathcal{C} \to \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$ is

(i) a map $F : Ob(\mathcal{C}) \to Ob(\mathcal{D})$;